

DYNAMIC INSTABILITY OF TWO-DIMENSIONAL NONDIVERGENT FLOW IN A BAROTROPIC ATMOSPHERE

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ABSTRACT

In discussing wave motions along a strong jet of westerlies, we can no longer neglect the variation of west wind with latitude, and therefore must consider the possibility of having a phase velocity whose value is between the maximum and minimum west-wind velocities. The study of wave motions is thus extended from the very long and slowly moving or retrograding waves into the realm of ordinary waves and cyclone waves and it is found that, for nondivergent barotropic motion, the condition for the presence of neutral and amplified waves with a phase velocity whose value is between the maximum and minimum wind velocity in the belt is the existence of *critical* points where the absolute vorticity has an extreme value. If no such point exists, then all perturbations must be damped.

When this condition is satisfied, both amplified (unstable) and neutral waves can be expected. The wave moving with a velocity equal to the current velocity at the critical point is neutral while those with a velocity less than this value but greater than the minimum wind velocity will be amplified. The amplification will be greatest when the phase velocity is intermediate between the latter two values; therefore, both fast and slowly moving waves will have little amplification. The degree of instability will also depend upon the sharpness of the velocity profile.

When the wave is unstable, the trough line will be directed from southeast toward northwest to the south of the point of minimum absolute vorticity and from southwest toward northeast to the north of the point of maximum vorticity.

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1. Introduction

In the last ten years the concept of conservation of absolute vorticity has been advanced in theoretical meteorology by many authors and also has been applied in synoptic meteorology. By introducing β , the variation of the Coriolis parameter $f = 2\omega \sin \phi$ with latitude, into the vorticity equation, Rossby (1939) discovered the law governing the motion of long waves in the upper atmosphere. Since all these studies are restricted to uniform westerlies, the waves are definitely neutral.

The study of dynamic instability may be dated back to Helmholtz and Reynolds (see Lamb, 1932) and since then has attracted the attention of many workers

in hydrodynamics and meteorology. Helmholtz investigated wave motions along a surface of discontinuity with an abrupt change in wind and density along the vertical, and showed that the common surface is unstable for sufficiently short wave disturbances. It was also shown that any finite wind discontinuity will have the effect of destabilization. Later on Rayleigh (1913) investigated the stability of horizontal parallel flows and tried to approximate the actual current profile by combining a number of belts, each with linear velocity distribution, and showed that the stability of the current will depend upon the shape of the profile. Heisenberg (1924) questioned the validity of this method; later on, most advancement of the study of instability of an inviscid fluid was made by Lin (1945; 1946), who also gave a physical interpretation in terms of conservation of vorticity.

Since all these studies are restricted to nonrotating systems, while in meteorology the earth's rotation plays a very important role in the dynamics of the atmosphere, the results of these investigations can hardly be applied in the field of meteorology except for some special phenomena occurring on a rather small scale, such as the formation of billow clouds. Parallel to this study of stability of wave disturbances was the consideration of the acceleration attained by a particle displaced from its original position, based largely on the assumption of conservation of momentum of the particle, which implies that there will be no change in the pressure field. This consideration was extended

into meteorology by Solberg (1936), and later on by Kleinschmidt (1941) and Van Mieghem (1944), by considering the balance of energy. The criterion thus obtained is that for instability to occur in two-dimensional horizontal motion or motion along isentropic surfaces, the wind shear along isentropic surfaces must be anticyclonic and with a value greater than the Coriolis parameter f ; all velocity profiles with cyclonic shear must be absolutely stable and the disturbances set in will be damped out, according to this theory. This instability can occur only to the south of a strong jet of westerlies. To the north, where we actually find our series of cyclones in the mean seasonal situation, the atmosphere is inertially stable. Perhaps we should distinguish the inertial instability discussed by Solberg from the instability we are considering, which for the case of wave disturbances, would depend upon the wave length. Furthermore, in view of the emphasis generally given to the theory of momentum transfer in turbulent motion, it seems very desirable to consider the problem from the point of view of vorticity transfer. Besides, most writers in dealing with the problem of instability of wave motion, were satisfied by showing that the wave velocity will become complex under certain conditions; very little attention has been given to the order of magnitude of the amplifying factor, owing to mathematical difficulties. But since we can rarely identify any atmospheric system in a time interval longer than one week, very small amplifying factors will be of no practical significance in meteorology.

The purpose of this investigation is to give a detailed study of the character of nondivergent horizontal wave motion in a barotropic atmosphere, without neglecting the change in the pressure field or the horizontal shear. A rather general distribution of the zonal wind will be assumed and a small perturbation superposed upon this basic flow, so that we can study the eventual changes of these small disturbances; this is essentially the problem of dynamic instability.

As for practical applications of this study, one would think that vertical motion and solenoidal fields as well as divergence and convergence will have deciding influence upon the development of the cyclones and anticyclones, so that it seems hardly justifiable to neglect these factors, and attribute all the instability to the distribution of zonal wind. Actually this is not the case. Although our discussions are mainly for two-dimensional nondivergent flows, similar results would be obtained for the very general case with vertical motion, solenoids, and divergence and convergence, if certain approximations are made in reducing the partial differential equation for a certain dependent variable to an ordinary differential equation. This equation is very similar to the vorticity equation of two-dimensional flow, except that the

coefficients of the equation involve quantities such as the vertical stratification, vertical wind shear, and the velocity of sound. Thus the only difference in the stability criterion will be a slight shift of the position of the critical point in the velocity profile and a slight difference of the degree of stability.

It should be pointed out that no attempt has been made to explain all the growth and decay of a very complicated system merely by dynamic instability. The dynamic stability or instability can only give some indication of the possibility for the development of some disturbances, while the actual mechanism of the disturbance itself must be studied from other considerations. Furthermore, since in some cases the general character of the basic current changes so rapidly that the disturbances can hardly be considered as small, and the system is receiving energy from outside, the dynamic stability will not give us the clue for these changes. But for many cases we actually find the main pattern of the flow remaining nearly the same for a long period of time without much change and with a series of waves, long or short, moving along it. It is to these cases that our results should apply; and if we can show that the order of magnitude of the amplifying factor is the same as in the actual cases, at least we can say that dynamic instability must be one of the main factors that are operating.

2. The perturbation equations

Since the large-scale atmospheric motions are mainly horizontal and the influence of compressibility negligible, we shall restrict ourselves to the study of purely horizontal and nondivergent motions. The atmosphere is assumed to be barotropic. Then the vorticity equation may be written in the following form:

$$\left(\frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) \nabla^2 \psi + (\beta - U'') \frac{\partial \psi}{\partial x} = 0, \quad (1)$$

where ψ is the perturbation stream function and $U = U(y)$ is the basic zonal velocity. If we put $\psi(x, y, t) = \Psi(y) e^{i\alpha(x-ct)}$, then

$$(U - c)(\Psi'' - \alpha^2 \Psi) + (\beta - U'')\Psi = 0, \quad (2)$$

where α is the wave number (a real quantity) and c , the phase velocity, may be complex; that is,

$$c = c_r + ic_i. \quad (3)$$

In (2), the primes denote differentiation with respect to y . If, in (3), $c_i \neq 0$, then the amplitude will be an exponential function of time t ; for $c_i > 0$, the amplitude will increase exponentially with time and the wave is said to be *amplified*; if, on the other hand, $c_i < 0$ then the amplitude will decrease with time and the wave will be *damped*. If $c_i = 0$, the amplitude will remain constant and the waves are said to be *neutral*.

Equation (2) is of second order and is to be solved under certain boundary conditions at $y = y_1$ and $y = y_2$. Since $U = U(y)$, we have a differential equation with variable coefficients. In the atmosphere, the velocity profiles of the westerlies generally have some symmetry with respect to some latitude; for simplicity, we shall discuss symmetric profiles only. In general, we may have two different kinds of profiles and therefore two different kinds of disturbances. The first is that for a limited belt: the disturbances disappear along its two boundaries $y = y_1$ and $y = y_2$, the velocity profile is symmetric with respect to the midpoint of the belt, as is shown in fig. 1. The second kind

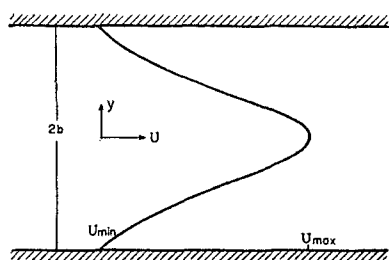


FIG. 1. The first type of symmetric velocity profile: limited belt of width $2b$.

is that for a belt extending to infinity in both directions. We shall assume in this case that there is a symmetric wind distribution in the central part and a constant wind outside this region, as is shown in fig. 2.

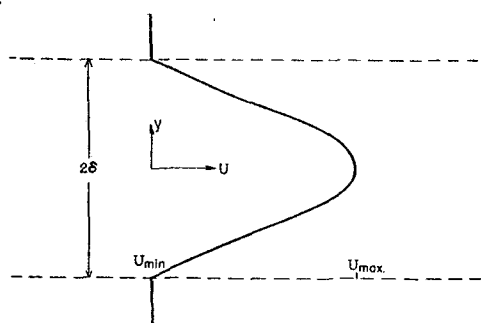


FIG. 2. The second type of symmetric velocity profile: uniform current velocity outside a belt of width 2δ .

Since only symmetric profiles and symmetric boundary conditions are considered, we should expect either symmetry or antisymmetry for the amplitude Ψ of the perturbation stream function. Therefore, only half of the belt need be considered. For the first kind of profile the conditions to be satisfied along the boundaries at $y = y_1$ and $y = y_2$ may be expressed as $\Psi = 0$. At the midpoint ($y = b$) of the region, we have $U' = 0$. Since it is very unlikely to have $v = 0$ along the line of maximum wind, we shall deal only with symmetric disturbances; for this type of disturbance $\Psi'(b) = 0$, since v has a maximum at $y = b$. Thus for the first type of profile the conditions to be satisfied are

$$\Psi(0) = 0, \quad \text{and} \quad \Psi'(b) = 0. \quad (4)$$

For the second type of profile, the same condition will be used for the midpoint, but a different condition must be applied at the two points where the central portion of the distribution joins the outer portions of uniform wind. Since (2) admits an exponential solution $\Psi \propto e^{\pm r y}$ for the region with constant wind, we should have $\Psi' \pm r\Psi = 0$ along the line where the profile changes to a constant wind profile. Thus if we take the origin of y along this line and the width of the belt 2δ , the conditions to be satisfied for the second type of profile are

$$\Psi'(0) - r\Psi(0) = 0, \quad \text{and} \quad \Psi'(\delta) = 0, \quad (5)$$

where $r^2 = \alpha^2 + \beta/(c - U_0)$. Our problem is now to solve equation (2) under the boundary conditions (4) or (5) and to find the necessary and sufficient conditions for having a complex c . This is a nonlinear characteristic-value problem with the parameter c .

Now the point $U = c$, where the wind velocity is equal to the phase velocity of the wave, is a singular point of (2). At this point, only one of the two fundamental solutions will remain regular while the other will have a logarithmically infinite derivative, corresponding to an infinite perturbation velocity u . Besides this, we shall have a sudden change of the imaginary part of u in crossing the singular point, which will give a sudden change of phase of u . In avoiding these difficulties, meteorologists have always made the assumption that c is always less than U ; but by so doing, we have already restricted ourselves to the special type of slowly moving or retrograding long waves, as will be seen later. Thus in order to remove the infinity of u and the sudden change of phase, we have to take the effect of lateral mixing into consideration. By introducing the frictional terms $\nu \nabla^2 \bar{u}$ and $\nu \nabla^2 \bar{v}$ into the equations of motion, the vorticity equation becomes

$$d\zeta/dt + \beta v = \nu \nabla^2 \zeta, \quad (6)$$

and after introducing the harmonic wave solution, it reduces to the following form

$$(U - c)(\Psi'' - \alpha^2 \Psi) + (\beta - U'')\Psi = -i\nu\alpha^{-1}(\Psi^{iv} - 2\alpha^2 \Psi'' + \alpha^4 \Psi). \quad (7)$$

In the free atmosphere, friction may be conceived in terms of lateral mixing, and the equation to be considered should be (7) instead of (2). When $U - c$ is not too small, the right-hand side of (7) can be neglected, so that we have the vorticity equation (2). But when U is nearly equal to c , this will no longer be the case and we must consider (7) instead of (2). Since $U - c = 0$ denotes only a point in the velocity profile and $U - c$ is very small only in the immediate vicinity of this singular point, the consideration of (7) is only necessary in a very thin belt around this point. At a small but finite distance from this singular point, the right-hand side of (7) becomes very small as compared

with the left, so that the solution must agree with that for (2). In this way (7) can be simplified enormously, and we get a correction for the solution of (2), so that the infinity of u can be removed and the value of the change of the imaginary part of Ψ and u determined. This method is called the method of *cross-substitution*.

3. Analytic properties of the stream function

Since (9) is of second order and the indicial equation is $r(r-1) = 0$, we see that only one of the two fundamental solutions remains regular at the singular point $y = y_s$. This solution is of the form

$$\Psi_1 = y + a_2 y^2 + a_3 y^3 + \cdots + a_n y^n + \cdots, \quad (8)$$

if we take the singular point as origin. The other linearly independent fundamental solution is of the form

$$\Psi_2 = b_0 + b_1 y + b_2 y^2 + \cdots + G_s \Psi_1 \ln y. \quad (9)$$

In (8) and (9), all the coefficients a_n , b_n , and G_s depend upon the velocity distribution $U(y)$, the wave number α and the phase velocity c . Our final aim is to find the values of α and the corresponding c that will make the general solution Ψ of (2) satisfy the given boundary conditions. Thus we see that our problem is really a problem of finding the characteristic values of α or c . This is done by substituting the general solution of (2), given by

$$\Psi = A\Psi_1 + B\Psi_2, \quad (10)$$

into the boundary conditions, and then eliminating the two arbitrary constants A and B . For the first type of profile and boundary conditions (4), the frequency equation will be of the form

$$F(\alpha, c) = \begin{vmatrix} \Psi_1(y_1; \alpha, c) & \Psi_2(y_1; \alpha, c) \\ \Psi_1'(y_2; \alpha, c) & \Psi_2'(y_2; \alpha, c) \end{vmatrix}, \quad (11)$$

where $y = y_1$ is the boundary and y_2 the midpoint. If we solve for c in terms of α and the other given parameters, we can find c_r and c_i . In general it is extremely difficult to find complex values of c satisfying (11). Frequency equations of the type (11) are generally solved by graphical methods for the real roots of c ; it is then inferred that we have complex c for values of α differing slightly from the critical value corresponding to real c . Thus we are actually looking for the critical values of α and the other parameters that will give a real c . But for many cases there is actually no real value for c and therefore this method will fail, for then we will not know whether c_i is positive or negative. Further, this method provides no information as to the magnitude of c_i . Therefore we shall employ another method to find all the possible real values for c , without actually solving (11), and then investigate c_i by a continuation.

Since the coefficients in Ψ_1 and Ψ_2 depend upon the velocity distribution $U(y)$, they cannot be given ex-

plicitly unless $U(y)$ is given. But what we really wish to know is only the general nature of the functions Ψ_1 and Ψ_2 (and of $\Psi = A\Psi_1 + B\Psi_2$) and their change of value in crossing the singular point y_s where $U = c$. For the latter purpose, a Taylor expansion of $U(y)$ around this point up to the third power of y will be sufficient. Thus if we take the origin at the singular point and take the positive direction toward the region where $U > c$, we shall have $U_s' > 0$ at the singular point, the subscript s signifying that the quantity is evaluated at the singular point. The coefficients in Ψ_1 and Ψ_2 can be determined by substituting $U(y)$ and the functions Ψ_1 , Ψ_2 given by (8) and (9) respectively into (2). For Ψ_2 , the constant terms reduce to $U_s' G_s + (\beta - U_s'') = 0$; therefore

$$G_s = (U_s'' - \beta)/U_s'.$$

We see that Ψ_1 remains regular for all finite values of y and the derivative of Ψ_2 becomes logarithmically infinite at $y = 0$, so that u becomes infinite at this point. The point $y = 0$ is a singular point for u and is also a branch point, where a sudden change of the imaginary part will occur.

Of course, u can never become infinite and there can never be a sudden change of phase in the real atmosphere. This can be explained by the presence of viscosity. Thus in the immediate vicinity of the singular point $y = 0$, we must take the action of viscosity into consideration by using (7) instead of (2) *i.e.*, using the method of cross-substitution. By this consideration, the infinity of u can be removed and the change of its phase made gradual. But since we are not particularly interested in the details within this very thin belt, we shall not discuss this method but only take the results that may be obtained from it, and consider that the solutions (8) and (9) are corrected such that they will not give an infinite u and a sudden change of phase for u in crossing the singular point.

For a real c , the change in the imaginary part of Ψ' in going from the positive to the negative side of $y = 0$ is determined by the method of cross-substitution and is given by $-G_s \pi i$.

When c is complex ($c = c_r + ic_i$), the coefficients of (2) become complex and there will be no singular point along the real axis of y . In this case we must consider the differential equation in the complex plane; U may also be considered as complex by an analytic continuation from the real axis. Then we shall have a singular point in the complex plane of y where we have $U = c$ and the two fundamental solutions of (2) are also given by (8) and (9) with the same coefficients, y being complex. If c_i is small, this singular point is at a distance c_i/U_s' from the real axis of y , above this axis for $c_i > 0$ and below it for $c_i < 0$, if we take U_s' as positive, where U_s' is the wind shear at the point $y = y_s$ corresponding to $U = c_r$ (fig. 3). For sufficiently

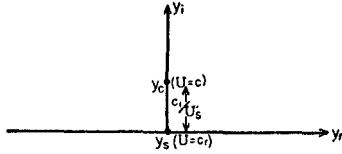


FIG. 3. The singular point y_0 in the complex plane (where $U = c$) in relation to the point y_s along the real y -axis where the phase velocity c_r is equal to the current velocity U .

small values of c_1 such that $|y| \gg |c_1/U_s'|$, there will also be a change of the imaginary part of Ψ' in passing from a positive real y to a negative real y , the main part of which is equal to $-G_s \pi i$, due to the logarithmic singularity.

4. Neutral waves

For the symmetric profiles we are considering on the rotating earth, the neutral waves can only have a phase velocity less than the maximum wind velocity. On the other hand the neutral waves may have a phase velocity less than the minimum wind velocity of the belt under consideration, which is impossible in a nonrotating system (Rayleigh, 1913). To show this, let us consider the solution of the differential equation (2), which may be written

$$\Psi'' = \left(\alpha^2 + \frac{\beta}{c - U} + \frac{U''}{U - c} \right) \Psi \equiv G\Psi,$$

as compared with the differential equation

$$F'' = \frac{U''}{U - c} F \equiv G_2 F. \quad (12)$$

Take the origin $y = 0$ at the boundary so that $\Psi(0) = 0$ and $F(0) = 0$, but $\Psi'(0) \neq 0$, $F'(0) \neq 0$. Then for $c > U_{\max}$, we have $G > G_2$; therefore, according to Sturm's comparison theorem, Ψ oscillates more slowly than F (Ince, 1944). But F has the solution

$$F = (c - U) \int_0^y \frac{dy}{(c - U)^2}, \quad (13)$$

which is always positive except at $y = 0$, where it is zero. Therefore Ψ can have no zeros for $y > 0$ and can not be zero at $y = 2b$, the other boundary; that is to say, the other boundary condition can never be satisfied by any value of c greater than the maximum wind velocity.

If, on the other hand, we assume $c < U_{\min}$, then G may be less than G_2 and there might be some values of c which makes Ψ equal to zero at $y = 2b$, so that Ψ is then the characteristic function which satisfies both of the two boundary conditions.

Now let us discuss the values of c . To do this, let us multiply (2) by F and (12) by Ψ , subtract and

integrate from $y = 0$ to $y = b$; we get

$$[\Psi'F - F'\Psi]_0^b = \int_0^b \left(\alpha^2 - \frac{\beta}{U - c} \right) F\Psi dy. \quad (14)$$

For Ψ , we have the conditions $\Psi(0) = 0$, $\Psi'(b) = 0$, $\Psi(b) > 0$, and $\Psi'(0) > 0$. If we take the solution of (12) to be

$$F = U - c, \quad (15)$$

which satisfies $F'(b) = 0$, (14) gives

$$c = \frac{\Psi'(0)U_{\min}}{\Psi'(0) + \alpha^2\gamma} + \frac{\bar{U} - \beta/\alpha^2}{1 + \Psi'(0)/\alpha^2\gamma}, \quad (16)$$

where $\gamma \equiv \int_0^b \Psi dy$ and $\bar{U}$ is the mean velocity defined by

$$\bar{U} = \gamma^{-1} \int_0^b U\Psi dy. \quad (17)$$

When $U = \text{constant}$ and Ψ is independent of y , then (16) gives

$$c = U - \beta/\alpha^2, \quad (18)$$

which is Rossby's equation.

When $U = \text{constant}$ and $\Psi = A \sin(\pi y/2b)$, then from (16) we have

$$c = U - \frac{\beta}{\alpha^2 + \pi^2/4b^2}, \quad (19)$$

which is Haurwitz' (1940) formula. Thus we see that Rossby's and Haurwitz' waves are the slowly progressive or retrograding long waves with a velocity less than the minimum wind velocity in the belt. This group of long and slowly moving waves does not give rise to any singularity to the vorticity equation (2). Thus the wave with a velocity c equal to the minimum wind velocity $U_{\min}$ is the demarcation between those waves which give a singularity to the vorticity equation and those which do not. We shall determine the wave length L_0 corresponding to this demarcation so that we can have some idea about the region of applicability of the results obtained from the assumption that $U_{\min} > c$. This demarcation will give the lower limit of the wave length for the waves studied by Rossby. By taking $U_{\min} = c$, (14) and (15) give

$$\beta \int_0^b \Psi dy = \alpha_0^2 \int_0^b (U - U_{\min})\Psi dy. \quad (20)$$

Therefore the corresponding shortest wave length is given by

$$L_0 = 2\pi\beta^{-1/2}(U - U_{\min})^{1/2}, \quad (21)$$

where $\bar{U}$ is defined as in (17). Take $\bar{U} - U_{\min} = 10 \text{ m sec}^{-1}$, which could not be too large as compared with the actual cases when the variation of U is great; then we find that L_0 is about 5200 km. Thus we see that the results obtained under the assumption of $U_{\min} > c$

can not be applied to the shorter waves such as the ordinary waves and cyclones that are associated with a strong jet of westerlies. Therefore, in continuing the long slowly moving waves to the ordinary and cyclone waves, we must consider the cases with $U_{\min} < c < U_{\max}$. It is in this region of phase velocity that we find the most prominent unstable waves, while the long and slowly moving waves are essentially neutral and can only have very small amplifications, because the most important contribution to the imaginary part of c comes from the change of value of Ψ in crossing the singular point (see appendix B). Thus the waves with a length larger than L_0 , given by (21), are essentially neutral.

Now let us consider the condition for the existence of neutral waves with $U_{\min} < c < U_{\max}$. We have seen in section 3 that in crossing the singular point where $U = c$, there is a change of phase for Ψ_2 equal to $-G_s \Psi_1 \pi i$, where $G_s = (U_s'' - \beta)/U_s'$, both for a real and complex c . Now we shall show that it is impossible for neutral waves (c real) to have such a change of phase.

Let us consider the general solution (10) of (2), given by

$$\Psi = A\Psi_1 + B\Psi_2,$$

where Ψ_1 and Ψ_2 are the solutions given in (8) and (9). If we separate the real and imaginary parts, we may write Ψ in the following form:

$$\Psi = \Psi_r + i\Psi_i, \quad (22)$$

with

$$\Psi_i'(0+) = \Psi_i'(0-) + G_s \pi. \quad (23)$$

In case of a real c , all the coefficients of (2) are real, so that both Ψ_r and Ψ_i must satisfy the differential equation and the boundary conditions separately, and therefore the Wronskian

$$\Delta = \Psi_i \Psi_r' - \Psi_i' \Psi_r$$

must be constant. But since both Ψ_r and Ψ_i must be zero on the boundaries and their derivatives must vanish at the midpoint of the belt, the Wronskian is equal to zero at all points except the singular point. At this point there is a sudden change of the Wronskian, given by

$$\Delta_+ - \Delta_- = \Psi_s^2 [\Psi_i'(0+) - \Psi_i'(0-)] = \Psi_s^2 G_s \pi,$$

which is impossible unless this quantity vanishes, since the Wronskian is equal to zero on both sides of the singular point. Thus for the neutral waves, this expression must be zero. From this we conclude that either $U_s'' = \beta$, *i.e.*, the singular point coincides with the point where the variation of the absolute vorticity with latitude is zero, or $\Psi_s = 0$. But the latter alternative is an impossibility unless $c = 0$ (or $c = U_{\min}$), *i.e.*, unless the singular point is at the boundary of the

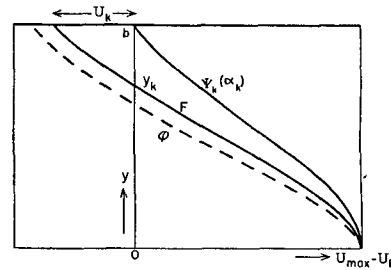


FIG. 4. The stream function Ψ_κ for the first velocity profile when there is a critical point y_κ .

belt, which belongs to the limiting case of the very slowly moving or retrograding waves. To show this, let us compare the solutions Ψ and F of (2) and (12); both are supposed to vanish at the singular point. Take the origin at the singular point and the nearer boundary at $y = y_0$. In this belt we have $G > G_2$, since $c - U > 0$; therefore Ψ will oscillate more slowly than F , according to Sturm's comparison theorem (Ince, 1944). Take $F = c - U$; then F will be positive from the singular point to the boundary (Ψ is also assumed to be positive near the singular point, *i.e.*, for y small). But since F is always positive except at the singular point, Ψ can not be zero at the boundary since it oscillates more slowly. Thus the only possibility of satisfying the boundary condition is that the singular point coincides with the boundary. The same argument can be applied to the other type of profile if we move the boundary to infinity.

Thus we have seen that the condition for having a neutral wave when $U_{\min} < c < U_{\max}$ is the existence of a critical point y_κ where $dZ/dy \equiv \beta - U'' = 0$. Now the question is: when this requirement is satisfied, can we actually have neutral waves, that is to say, can the given boundary conditions be satisfied by some real values of c ? It can be shown that if $dZ/dy \equiv \beta - U'' = 0$ at some point y_κ , then we actually have the characteristic values $c = U_\kappa$ (or $\alpha = \alpha_\kappa$) and that it is the only one for the problem. To prove this let us consider the Sturm-Liouville system

$$\Psi'' = \left(\alpha^2 + \frac{U'' - \beta}{U - U_\kappa} \right) \Psi; \quad \Psi'(0) = 0, \quad (24)$$

where the midpoint of the velocity profile has been taken as the origin. Let us suppose that our profile is such that the critical point is a point where the absolute vorticity has its extreme value, *i.e.*, $U'' - \beta$ changes its sign there; then $(U'' - \beta)/(U - U_\kappa)$ will be negative in the whole region $(0, b)$.

If we take $\alpha = 0$ and let Ψ take the value $U(0) - U_\kappa > 0$ at the midpoint, then it will have a zero at a point $y < y_\kappa$, for the solution $F = U - U_\kappa$ of the system

$$F'' = \frac{U''}{U - U_\kappa} F; \quad F'(0) = 0, \quad (25)$$

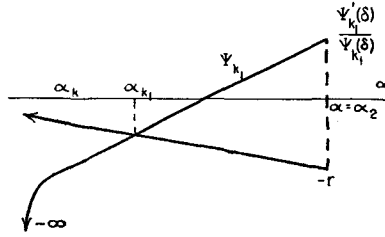


FIG. 5. The stream function Ψ_k and the wave number α_k for the second velocity profile.

has a zero at $y = y_k$ and $(U'' - \beta)/(U - U_k) < U''/(U - U_k)$ in the interval $(0, y_k)$; therefore, Ψ oscillates more rapidly than F and the first zero of Ψ will be nearer to the origin than the first zero of F , since both functions start from the same value at $y = 0$ (fig. 4). Thus Ψ will be negative at $y = b$. If we let α increase from zero, this zero point of Ψ will move toward the boundary $y = b$, and finally, when $\alpha = \alpha_k$, we shall have the zero point of Ψ exactly at $y = b$, i.e., the other boundary condition is satisfied by this particular value α_k , the characteristic number, and Ψ_k is then the characteristic function.

For the second profile, the zero point of Ψ is shifted to infinity, but it is easily shown that the characteristic value α_{k1} remains finite, for if we take $\alpha = \alpha_2$ greater than the upper limit of $(U'' - \beta)/(U - U_k)$, so that the lower bound of $G_2 \equiv \alpha_2^2 + (U'' - \beta)/(U - U_k)$ is k^2 , which is positive, then the system

$$\varphi'' = k^2 \varphi; \quad \varphi'(0) = 0 \quad (26)$$

has the non-oscillatory solution $\varphi = A \cosh ky$. When α decreases from α_2 to α_k , the characteristic value for the first profile, a zero of Ψ will appear at $y = \delta$. Thus there must be a value $\alpha = \alpha_{k1}$ in the interval $\alpha_2 > \alpha_{k1} > \alpha_k$, which will make Ψ equal to zero at infinity but not at any finite distance; this is the characteristic value for the second profile. This can also be proved by applying Sturm's oscillation theorem to (24) and (26), and finding the characteristic value $\alpha = \alpha_{k1}$ that makes Ψ , the characteristic function, satisfy the other condition

$$\Psi'(\delta) + r\Psi(\delta) = 0,$$

where $r = [\alpha^2 + \beta/(c - U_0)]^{1/2} > 0$. We have seen that, when $\alpha = \alpha_2$, $\Psi'/\Psi = \varphi'/\varphi > 0$, while as $\alpha \rightarrow \alpha_k$, $\Psi(\delta)$ approaches zero from above, i.e., $\Psi'(\delta) < 0$ and $\Psi'(\delta)/\Psi(\delta) \rightarrow -\infty$. Since $-r$ increases from $-[\alpha_2^2 + \beta/(c - U_0)]^{1/2}$ to $-[\alpha_k^2 + \beta/(c - U_0)]^{1/2}$ as α decreases from α_2 to α_k , while $\Psi'(\delta)/\Psi(\delta)$ decreases from a positive value to $-\infty$, there must be a value of $\alpha = \alpha_{k1}$, for which we have $\Psi'(\delta)/\Psi(\delta) = -r$; therefore α_{k1} is the characteristic value of the problem, and its value is greater than α_k (see fig. 5).

Thus, if there is only one critical point in the profile, we shall have one and only one neutral wave, with $c = U_k$ and $L = L_k$; all other waves with a phase velocity

differing from U_k will be either amplified (unstable) or damped, aside from the group of slowly moving or retrograding long waves with $c < U_{\min}$. Therefore we must conclude that all the waves propagating with a velocity less than U_k but greater than $U_{\min}$ will be either unstable or damped, and it will be shown later that they are actually unstable waves.¹

If the condition $dZ/dy \equiv \beta - U'' = 0$ is satisfied for all y in the belt, which means that the profile is made up of constant-vorticity profiles, then the waves may move with any current velocity in the belt and are all neutral. This is obvious since then (2) reduces to $(U - c)\nabla^2\Psi = 0$, so that $U = c$ is always possible. In this case, there is a linear relationship between c and L .

5. The necessary condition for occurrence of unstable waves

Having now found all the neutral waves, let us consider the solutions with complex c . Equation (2) is to be solved under the condition that $\Psi = 0$ at the two boundaries $y = y_1$ and $y = y_2$. Then we have the following Sturm-Liouville system

$$\frac{d^2\Psi}{dy^2} - \left(\alpha^2 + \frac{U'' - \beta}{U - c} \right) \Psi = 0, \quad (27)$$

$$\Psi(y_1) = 0, \quad \Psi(y_2) = 0.$$

We shall investigate the conditions under which we may have a complex $c = c_r + ic_i$. Suppose c complex and write $(U - c)^{-1}$ in the form $g_1 + ig_2$, with

$$g_1 = \frac{U - c_r}{(U - c_r)^2 + c_i^2}, \quad g_2 = \frac{c_i}{(U - c_r)^2 + c_i^2}. \quad (28)$$

In this case the coefficient of Ψ in (27) becomes complex and we must take Ψ to be complex, such that $\Psi = \Psi_r + i\Psi_i$. The system (27) then becomes, after separating the real and imaginary parts

$$\begin{aligned} \Psi_r'' - [\alpha^2 + (U'' - \beta)g_1]\Psi_r + (U'' - \beta)g_2\Psi_i &= 0, \\ \Psi_i'' - [\alpha^2 + (U'' - \beta)g_1]\Psi_i - (U'' - \beta)g_2\Psi_r &= 0, \end{aligned}$$

with the conditions $\Psi_r(y_1) = \Psi_r(y_2) = 0$, $\Psi_i(y_1) = \Psi_i(y_2) = 0$. If we multiply the first equation by Ψ_i

¹ It may be mentioned that Thompson's (1948) solution is actually the one discrete neutral solution we are considering in the range of velocity $U_{\min} < c < U_{\max}$ or the one slowly moving wave with $c < U_{\min}$. His equation (4) is an equation defining the zonal current (for finite-amplitude motions we may take the values along a trough where $v = 0$) as a sine-curve profile with the parameter c . If we differentiate it with respect to y , we get the condition

$$Z' \equiv \beta - U'' = -k^2\Psi' = k^2(U - c),$$

where $Z \equiv f + \nabla^2\Psi$ is the absolute vorticity. Thus c can not take the value U unless $dZ/dy = 0$ at that point (which is the critical point). If no such point exists in the profile, c must be different from all values of U in the belt. Thus, his equation (15) must not be interpreted as a continuous relation between c and L in this range of velocity, but only expresses c_k in terms of L_k , which is determined from the critical point.

and the second by Ψ_r and subtract, we get

$$\frac{d}{dy} \left(\Psi_i \frac{d\Psi_r}{dy} - \Psi_r \frac{d\Psi_i}{dy} \right) = - (U'' - \beta) g_2 (\Psi_r^2 + \Psi_i^2).$$

Now let us integrate this from y_1 to y_2 , and make use of the boundary conditions; thus,

$$\int_{y_1}^{y_2} (\beta - U'') g_2 (\Psi_r^2 + \Psi_i^2) dy = 0. \quad (29)$$

But since g_2 is either always positive or always negative, according to the sign of c_i , and $\Psi_r^2 + \Psi_i^2$ always positive, we can have the integral equal to zero only if $dZ/dy \equiv \beta - U''$ is identically zero or if $\beta - U''$ somewhere changes its sign within the region, *i.e.*, it must be zero somewhere in the region.

Thus the necessary condition for having a complex c is that

$$U'' - \beta = 0 \quad (30)$$

at some point $y = y_k$ in the region, and changes its sign there. This condition is both necessary and sufficient for c_i to be different from zero. It is necessary as the proof stands. By sufficient we mean that, whenever $U'' - \beta = 0$ at some point in the region, and changes its sign there, then we must have a complex c with $c_i \neq 0$. This is rather obvious since we have only a single isolated stable solution with $c = U_k$, $L = L_k$, while the frequency equation (11) will give c as a function of α or L for every value of L ; therefore, for L different from L_k , we must have $c_i \neq 0$.

The sufficiency of this condition for the existence of amplified waves will be proved in the next section by a more direct and rigorous method by demonstrating (1) the existence of characteristic values of c and α in the vicinity of the neutral wave (c_k , L_k) such that $c_i \neq 0$, and (2) that the sign of c_i is positive for $L > L_k$. It must be pointed out that the condition (30) is only necessary for neutral and amplified waves but not for damped waves since the solution is not valid all along the real axis when $c_i < 0$ (see Lin, 1945).

The same criterion can also be applied to motions over the spherical earth. In this case, the vorticity equation can be written in the form

$$(\lambda - \mu) \left(\frac{d^2 \Psi}{dy^2} - n^2 \Psi \right) - \frac{dZ}{dy} \Psi = 0, \quad (31)$$

where $y = \ln(\sec \phi + \tan \phi)$, λ is the relative angular velocity of the current, μ the angular velocity of the disturbance, and $\psi = \Psi(y)e^{in(\lambda - \mu)t}$ the stream function for the perturbation, defined by

$$\phi \cos \phi = \partial \psi / \partial \lambda, \quad \lambda \cos \phi = - \partial \psi / \partial \phi.$$

Since equation (31) is of the same form as (2) and the boundary conditions may be taken as $\Psi = 0$ at $y = 0$ (the equator) and $y = \infty$ (the pole), the same conclusion will be obtained as before, *i.e.*, the absolute

vorticity must have some maximum or minimum in the region. Thus the curvature of the earth does not have much effect on the nature and stability of the disturbances.

The physical explanation of this condition for amplified waves may be sought from the meaning of the condition $dZ/dy \equiv \beta - U'' = 0$, which represents the points with an extreme value of absolute vorticity. If we take the zonal distribution of wind as a distribution of vortex filaments, and calculate the acceleration attained by a vortex element displaced from its original position in such a field, we shall find that the transversal acceleration will be given by (Lin, 1945)

$$\bar{a}_y = \Gamma^{-1} \iint v^2 \frac{dZ}{dy} dx dy,$$

where $\Gamma \equiv \oint \zeta dx dy$ = circulation of the element, and y will be replaced by the expression $\ln(\sec \phi + \tan \phi)$ in case of motion over the spherical earth. Thus $\bar{a}_y$ is proportional to $dZ/dy \equiv \beta - U''$, and is always directed toward the original position if $\beta - U''$ is of the same sign in the entire region. Only when there are some points where $\beta - U''$ changes sign, could the elements gain an acceleration in the direction away from their original positions, and only those elements that actually have crossed these points will be thus accelerated.² It should be mentioned that the fulfillment of the condition $U'' = \beta$ only gives the *possibility* for instability to occur, unlike the criterion obtained from the assumption of conservation of absolute momentum which asserts that the situation will be definitely unstable for all moving elements in the region where we have $U' > f$, regardless of the character or size of the disturbance. The selective character of the new criterion is in agreement with results obtained from the study of stability of wave motions, which shows that the condition of stability will depend upon the wave length of the disturbance.

If, on the other hand, we have a vortex superimposed in the current of basic flow, it will also be subject to a force $\bar{a}_y$ per unit mass. Thus, if in some locality and at some instant, the motion deviates from purely horizontal nondivergent barotropic conditions, and a concentration of cyclonic or anticyclonic vorticity is produced through any process in the atmosphere, such as convergence or divergence, vertical motion or the action of a solenoidal field, while the motion of the surrounding atmosphere still retains its

² This selective nature of the stability criterion may be explained by the nature of the field of force. In the case of vertical stratification or horizontal distribution of centrifugal force, the force is directed in one direction only, while in our case, the point where $U'' = \beta$ may be considered as the center of a repelling or attracting force, acting in opposite directions on its two sides. Thus for small displacements, the element will be acted on by a restoring force, driving it back to its original position; but if the displacement is great enough such that the element reaches the other side of this center of force, an acceleration away from its original position will result.

horizontal and nondivergent character, then this vortex will be moving under a force $\bar{a}_y$ in the north-south direction. In the ordinary case of a strong jet of westerlies the absolute vorticity has a maximum to the north of the latitude of maximum wind and a minimum to the south. Thus if a cyclonic vortex is produced to the north of the point of maximum absolute vorticity, the force will be directed toward the south. The effect of the motion between the latitudes where Z has its extreme values gives a force in the opposite direction. It is the resultant of these two opposing forces that determines the motion of the vortex. The southward motion and the cutting-off of the lows may thus be explained by the interaction between the distribution of absolute vorticity in the basic flow and the velocity field due to the superimposed vortex.

The concentration of vorticity may be expected to occur in the troughs and ridges of the upper waves, especially when the wave length corresponds to the maximum unstable condition.

6. Amplified waves

We have proved in the preceding sections that when we have one critical point $y = y_*$ where $dZ/dy \equiv \beta - U'' = 0$, we have one and only one neutral wave with a real phase velocity and that the value of this phase velocity is $c = U_*$; for all other velocities such that $c_r \neq U_*$, we shall have $c_i \neq 0$.

This statement is proved by considering the solution $\Psi = \Psi_* + (d\Psi/d\lambda)\Delta\lambda$ corresponding to $c = c_* + (dc/d\lambda)\Delta\lambda$, slightly different from the neutral solution Ψ_* corresponding to $c = c_* = U_*$, $\lambda = \lambda_* \equiv \alpha_*^2$. If we write $\Delta c = \Delta c_r + i c_i$, it will be found that c_i and Δc_r are given by

$$c_i = -D\pi E^{-1}\Delta c_r, \quad (32)$$

$$(1 + D^2\pi^2 E^{-2})\Delta c_r = -\frac{\Delta\lambda}{E} \int_0^b \Psi_*^2 dy, \quad (33)$$

where

$$E \equiv \lim_{\epsilon \rightarrow 0} \left(\int_0^{y_*-\epsilon} + \int_{y_*+\epsilon}^b \right) \frac{U'' - \beta}{(U - c_*)^2} \Psi_*^2 dy; \quad (34)$$

and $D \equiv (U_*''' - \beta')/U_*'^2$. Since in general we have $U'' > \beta$ from 0 to y_* (origin at the boundary) and $U'' < \beta$ from y_* to b , we have $D < 0$. Thus if Δc_r and E have the same sign, we shall have $c_i > 0$, while if they have opposite signs, we shall have $c_i < 0$. Since β is positive, the absolute value of $U'' - \beta$ is larger when U'' is negative than when U'' is positive; and since Ψ_* has its maximum value at $y = b$, we should expect E to be negative by an inspection of (34); this is also shown to be the case in the examples discussed in section 7. Thus in order to have $c_i > 0$, we must have $\Delta c_r < 0$ and (33) shows that we must have $\Delta\lambda \equiv \alpha^2 - \alpha_*^2 < 0$, i.e., the waves longer than the neutral wave with $c = U_*$ and $L = L_*$ are unstable. If

$E \rightarrow 0$, then $\Delta c_r/E$ will be given by $-\Delta\lambda \int_0^b \Psi_*^2 dy$, which is positive for $\Delta\lambda < 0$, that is to say, c_i is also positive for $\Delta\lambda < 0$. If E is positive, a positive c_i will also require $\Delta\lambda < 0$. Thus we see that unstable waves in the neighborhood of the neutral wave with $c = U_*$ always have longer wave length than L_* . The detailed procedure in obtaining (32) and (33) is given in appendix A.

These results, with slight modification, are valid also for the second profile, as indicated in appendix A.

In the neighborhood of the stationary (or $c = U_{\min}$) neutral solution, it is demonstrated in appendix B that, for $\Delta c_r > 0$ (i.e., for $c_r > U_{\min}$), c_r and c_i are given by the approximate equations

$$c_r \left(U'(0) + \alpha_0^2 \int_0^b U dy \right) = \Delta(\alpha^2) \int_0^b U^2 dy, \quad (35)$$

$$c_i \left(U'(0) + \alpha_0^2 \int_0^b U dy \right) = G_s \pi c_r^2, \quad (36)$$

which show that c_i is always positive since $U_0'' - \beta > 0$ and $U_0' > 0$.

From the preceding discussions we see that all the waves with $U_{\min} < c_r < U_*$ are amplified and those with $c_r > U_*$ are damped. Thus, if we draw an L , c -diagram, we shall have three regions separated from each other by the values $L = L_0$ and $L = L_*$ corresponding to the phase velocities $c = U_{\min}$ and $c = U_*$. To the right of $L = L_0$, the waves are essentially neutral; in the region $L_0 > L > L_*$ ($U_{\min} < c_r < U_*$) we have $c_i > 0$ and the waves are amplified. The degree of instability will be greatest somewhere in the central part of this region. To the left, with $L < L_*$ and $c_r > U_*$, the waves will be damped (see fig. 6).

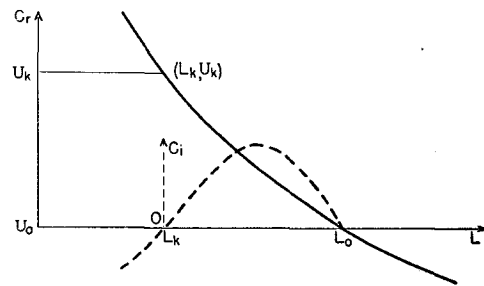


FIG. 6. Schematic representation of c_r and c_i as functions of wave length.

We may now discuss the direction of the troughs and ridges. For neutral waves, since there can be no change of phase, the troughs and ridges must be in the direction from south to north. For unstable waves, with $U'' \neq \beta$ at the point where $U = c_r$, the trough lines will be tilted, in the direction from southeast toward northwest to the south of the point of minimum absolute vorticity, and from southwest toward northeast to the north of the point maximum absolute vorticity. This can be proved by the following considerations.

Along the trough, we have $v = 0$, where v is the real part of $\partial\psi/\partial x$. Since $\psi = \Psi e^{ik(x-ct)}$, where $\Psi = \Psi_r + i\Psi_i$ and $c = c_r + ic_i$, we find that

$$v = -k(\Psi_r \sin k\xi + \Psi_i \cos k\xi)e^{c_i t}, \quad (37)$$

in which $\xi = x - c_r t$. The trough lines are therefore to be determined from

$$\tan k\xi = -\Psi_i/\Psi_r.$$

Since the direction of the trough lines is determined by the sign of ξ' , we differentiate the above equation with respect to y :

$$k\xi' \sec^2 k\xi = (\Psi_r' \Psi_i - \Psi_i' \Psi_r)/\Psi_r^2. \quad (38)$$

If we take y to be directed from the region where $U < c_r$ to the region where $U > c_r$, then for slightly amplified waves, the quantity on the right of (38) is negative, according to Tollmien (1929). Thus the troughs and ridges are directed from southeast toward northwest to the south of the point of minimum Z , and from southwest toward northeast to the north of the point of maximum Z .

This conclusion is in agreement with the consideration of the change of the kinetic energy of the eddying motion with time. Thus, the average value over a wave length of the correlation between the perturbation velocities u and v is given by

$$\overline{uv} = \frac{1}{2}k(\Psi_r' \Psi_i - \Psi_i' \Psi_r),$$

which, as already noted, is negative in the region where $U' \equiv \partial U/\partial y > 0$. Hence $U'\overline{uv}$ is negative, and the kinetic energy of the perturbation increases, while that of the basic flow decreases (see Lamb, 1932, pp. 676-677). For damped waves, the direction of the trough lines will be in the opposite sense and the correlation between u and v will be positive, corresponding to a decrease of the kinetic energy of the eddying motion and an increase of the kinetic energy of the basic flow.

7. Examples with sine-curve velocity profile

Let us apply the foregoing discussions to the sine-curve velocity profiles. For this kind of periodic profile, the stream function Ψ_k for the neutral wave with $c = U_k$ can be obtained explicitly in a simple form. Since we can fit a sine curve to most of the observed wind distributions by varying the parameters such as the maximum wind velocity $U_{\max}$, the width of the belt $2b$, and the position of the critical point $y = y_k$, we shall discuss this profile in some detail.

A. Limited belt.—For the first profile, we may write

$$U = U_k + \frac{\sin k(y - y_k)}{1 + \sin ky_k} U_{\max} + \frac{\beta}{k^2}, \quad (39)$$

which satisfies the conditions that $U = 0$ at $y = 0$ and

$U''' = \beta$ at $y = y_k$. In (39), we have $k = \frac{1}{2}\pi/(b - y_k)$; y_k is the inflection point of the profile (where $U'' = 0$), U_k is the current velocity at this point, and the mid-point of the profile is at $y = b$, where the wind velocity is equal to $U_{\max}$.

If we substitute (39) into (2) and put $c = U_k$, we get the equation for Ψ_k :

$$\Psi_k'' = (\alpha_k^2 - k^2)\Psi_k, \quad (40)$$

which admits the solution $\Psi_k = B \sin py$, where

$$p^2 = k^2 - \alpha_k^2. \quad (41)$$

The boundary conditions $\Psi_k(0) = 0$, $\Psi'(b) = 0$ are satisfied by $p = (2n + 1)\pi/2b$, where n is any integer. But since $0 \leq y_k < b/2$, the only possible value of p is $\pi/2b$.³ Therefore we get

$$\Psi_k = \frac{\sin(\pi y/2b)}{\sin(\pi y_k/2b)}, \quad (42)$$

the constant factor being chosen to make $\Psi_k = 1$ at the critical point.

The wave number α_k is given by

$$\alpha_k^2 = \frac{\pi^2}{4b^2} \left[\left(1 - \frac{y_k}{b} \right)^{-1} - 1 \right]. \quad (43)$$

Thus the stable wave with $c_k = U_k$ is completely determined by the value of y_k , which depends upon y_k , and the stream function is given by (42) while the wave length $L_k = 2\pi/\alpha_k$ is determined by (43).

For $y_k = 0.333b$, we have $y_k = 0.245b$ and

$$\alpha_k b = 0.559\pi, \quad c_k = U_k = 0.414U_{\max} - 0.181b^2\beta. \quad (44)$$

For $y_k = 0.25b$, we have $y_k = 0.139b$ and

$$\alpha_k b = 0.440\pi, \quad c_k = U_k = 0.333U_{\max} - 0.226b^2\beta. \quad (45)$$

Thus we see that the greater the value of c_k , the shorter will be the wave length L_k corresponding to a larger α_k . Since α_k is inversely proportional the width of the belt $2b$, L_k is directly proportional to b . For $y_k = b/3$ and $2b = 20$ degrees latitude, we get $L_k = 36$ degrees latitude.

The amplifying oscillations in the neighborhood of the stable solution Ψ_k are calculated by using (32) and (33), and the values in the vicinity of $c = 0$ (or $c = U_{\min}$) are calculated by using (35) and (36). Since we have only considered the first term in the Taylor expansion of c , the formulas thus obtained will hold

³ If we eliminate the terms involving y from (39) and its second derivative and then integrate from 0 to $2b$, we get, by using (41):

$$c_k = U_k = \bar{U} + (\bar{U}'' - \beta) \left(\frac{\pi^2}{4b^2} + \frac{4\pi^2}{L_k^2} \right)^{-1},$$

which relates the phase velocity c_k to the average value of the current $\bar{U}$; $\bar{U}''$ is the difference of the shear at the two boundaries divided by $2b$, the wave length is L_k , and the breadth of the belt $2b$. This equation looks like a generalized Rossby's equation, but it holds only for the neutral wave with c_k and L_k , and must not be interpreted as a continuous relation between c and L .

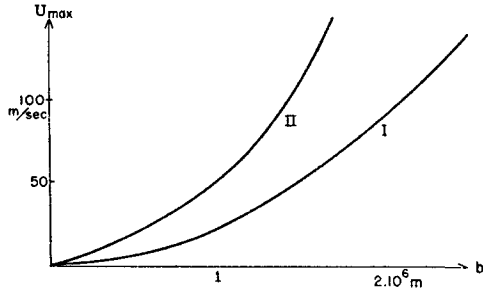


FIG. 7. The sharpness of the velocity profiles expressed as a relation between the maximum current velocity $U_{\max}$ and the width of the belt, b .

only for small values of $\Delta\lambda = \Delta(\alpha^2)$. The values for the intermediate regions between $\alpha^2 = \alpha_0^2$ and $\alpha^2 = \alpha_k^2$ are obtained by interpolation. If more accurate values are desired, we must calculate the second and higher derivatives; but since we are concerned only with orders of magnitude, no attempt has been made in this direction. Now let us consider the different cases separately.

Case I. Weak jet. Let $U_{\max} = 2.2 \times 10^{-11} b^2 \text{ m sec}^{-1}$ (fig. 7, curve I). With $y_t = b/3$, $y_k = 0.2447b$, the values of c_k and α_k are given by (44), which, in this case, reduce to

$$c_k = 0.293 U_{\max}, \quad \alpha_k b = 0.559\pi. \quad (46)$$

In the vicinity of $c = c_k$, $\alpha = \alpha_k$, we have, from (33) and (32),

$$c_r/U_{\max} = 0.099 + 0.0634\alpha^2 b, \quad (47)$$

$$c_i/U_{\max} = 0.044 - 0.0144\alpha^2 b. \quad (48)$$

From (47) and (48) we see that for $\alpha < \alpha_k$, we have $c_r < c_k$ and $c_i > 0$; and for $\alpha > \alpha_k$, we have $c_r > c_k$, $c_i < 0$. Thus waves longer than the neutral wave $c = U_k$ are unstable and slower while the shorter waves are faster and damped. In the vicinity of $c = 0$ (or $= U_{\min}$), the corresponding values of c_r and c_i are obtained by using (35) and (36). Here we have $\alpha_0 b = 0.94$ and

$$c_r/U_{\max} = 0.312\alpha^2 b^2 - 0.275, \quad (49)$$

$$c_i/U_{\max} = 3.62 (c_r/U_{\max})^2. \quad (50)$$

The stream function Ψ_0 is shown in fig. 8 together with

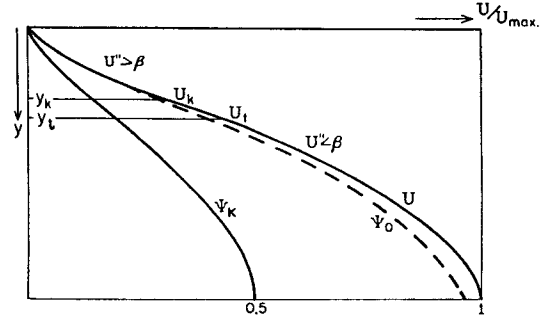


FIG. 8. $\Psi_k(y)$ and $\Psi_0(y)$ for the first velocity profile in comparison with $U(y)$ as functions of y . Note that $\Psi_0(y)$ is nearly equal to $U(y)$.

U and Ψ_k , for comparison. The values of c_r and c_i , given by (47), (48), (49), and (50) are plotted in fig. 9(A) for $y_t = b/3$. The central parts are obtained by interpolation.

For the case with $y_t = b/4$, we have $y_k = 0.1388b$ and

$$c_k = 0.179 U_{\max}, \quad \alpha_k b = 0.442\pi, \quad \alpha_0 b = 0.910; \quad (51)$$

the computed values of c_r and c_i are plotted in fig. 9(B). We see that the values of c_k and c_i have been diminished enormously as compared with the case for $y_t = b/3$. Although the condition that $Z \equiv f - U'$ has an extreme value at some points is sufficient for the existence of some $c_i > 0$, the value of c_i depends upon the position of the critical point y_k , which determines the factor $D \equiv U_k'''/U_k'^2$. The above examples show this clearly. With the point y_k shifted toward the boundary $y = 0$, where the disturbance is supposed to be zero, very few particles can actually get across this point and the disturbance is therefore comparatively stable.

Case II. Strong jet. Let $U_{\max} = 5 \times 10^{-11} b^2 \text{ m sec}^{-1}$ (fig. 7, curve II). With $y_t = b/3$, we have $y_k = 0.294b$ and

$$c_k = 0.360 U_{\max}, \quad \alpha_k b = 0.559\pi, \quad \alpha_0 b = 0.626. \quad (52)$$

The profile is represented by the same curve in fig. 7, but now the ratio $b^2/U_{\max}$ has decreased so that the profile is sharper than that for case I. The value of U_k has increased from $0.293 U_{\max}$ to $0.360 U_{\max}$, and y_k is now nearer to y_t . The values of c_r and c_i for this case are represented in fig. 9(C).

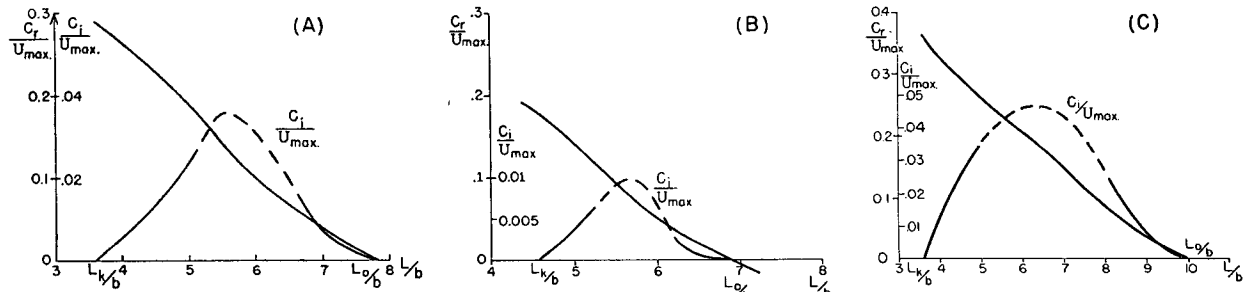


FIG. 9. c_r and c_i for the first sine-curve velocity profiles. A: for the weaker jet (I in fig. 7) with $y_t = b/3$; B: for the weaker profile (I in fig. 7) with $y_t = b/4$; C: for the stronger jet (II in fig. 7) with $y_t = b/3$.

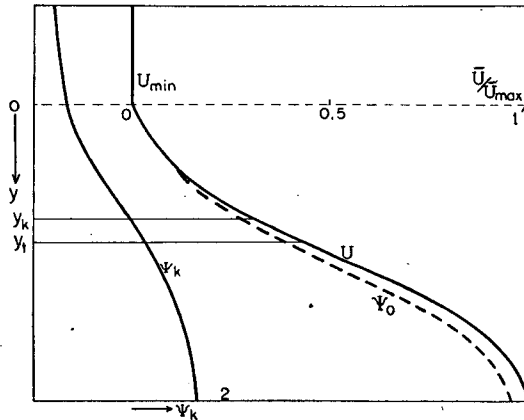


FIG. 10. Ψ_k and Ψ_0 for the second velocity profile. Note that Ψ_0 is nearly equal to $U(y)$.

From these examples we see that the effect of sharpening the profile is to increase the values of c_k , c_r , and c_i , i.e., increase the instability.

B. Belt extending to infinity.—For the second type of profile, equations (32) and (33) are to be used for the values near $c = U_k$, $\alpha = \alpha_k$, with the modification mentioned at the end of section 7, while the values in the vicinity of $c = U_{\min} = U_0$ are given by the same expressions as in the preceding example for a limited belt. For this case, the conditions to be satisfied are $U = U_0$ for $y = 0$, and $U'' = \beta$ at $y = y_k$. These give the following expression for U :

$$U = U_k + \frac{\sin k(y - y_t)}{1 + \sin ky_t} \bar{U}_{\max} + \frac{\beta}{k^2}, \quad (53)$$

the same as (39) except with $\bar{U}_{\max} = U_{\max} - U_0$, $k = \frac{1}{2}\pi/(\delta - y_t)$ (the midpoint is $y = \delta$, where $U = U_{\max}$). Let us use $\bar{c}_k \equiv c_k - U_0 = U_k - U_0$; then the equation for Ψ_{k1} reduces to (40), as in the preceding example, but now the conditions to be satisfied are

$$\begin{aligned} \Psi_{k1}'(0) &= r\Psi_{k1}(0); \quad r^2 = \alpha_k^2 + \frac{\beta}{U_k - U_0}; \\ \Psi_{k1}'(\delta) &= 0. \end{aligned} \quad (54)$$

The solution which satisfies the second condition is, when normalized

$$\Psi_{k1} = \cos p(y - \delta)/\cos p(y_k - \delta),$$

so the first condition requires the relation

$$r = p \tan p\delta. \quad (55)$$

Thus we get

$$\frac{p}{\cos p\delta} = \left(k^2 + \frac{\beta}{U_k - U_0} \right)^{\frac{1}{2}} = (r^2 + p^2)^{\frac{1}{2}}, \quad (56)$$

with $0 \leq y_t < \delta/2$.

Case I: Weak jet. Let $y_t = \delta/3$, $y_k = 0.245\delta$. Then

$$\begin{aligned} p\delta &= 0.364\pi, \quad \alpha_k\delta = 2.055, \quad r\delta = 2.56 \\ \bar{c}_k &= U_k - U_0 = 0.293\bar{U}_{\max}, \quad \alpha_0\delta = 0.94. \end{aligned} \quad (57)$$

The stream functions for the central and outer parts respectively are plotted in fig. 10, and the values of c_r and c_i are plotted in fig. 11(A). For a belt with $2\delta = 25$ degrees latitude, the wave length L_k for the stable wave with $c = U_k$ is equal to 39 degrees latitude while the wave with $c = U_0$ has the length $L_0 = 83$ degrees latitude.

Case II: Strong jet. Let $\bar{U}_{\max} = 5 \times 10^{-11} \delta^2 \text{ m sec}^{-1}$. For $y_t = \delta/3$ we have $y_k = 0.294\delta$ and

$$\begin{aligned} \bar{c}_k &= 0.360\bar{U}_{\max}, \quad \alpha_k\delta = 2.07, \\ p\delta &= 0.354\pi, \quad r\delta = 2.32, \quad \alpha_0\delta = 0.63. \end{aligned} \quad (58)$$

From these we get

$$\bar{c}_r/\bar{U}_{\max} = 0.113 + 0.058\alpha^2\delta^2, \quad (59)$$

$$c_i/\bar{U}_{\max} = 0.228 - 0.053\alpha^2\delta^2. \quad (60)$$

Equations (59) and (60) are plotted in fig. 11(B), together with the values for c_{r0} and c_{i0} , given by (49) and (50). From these examples we see that the profile of unlimited lateral extent is comparatively more unstable than that for a limited belt. Here the destabilizing effect of sharpening the profile can also be seen very clearly with the increase of c_r and c_i .

From the foregoing examples we see that for the unstable waves with $L_k < L < L_0$ and $U_k > c_r > U_0$, the amplification factor αc_i is of the order of 10^{-5} sec^{-1} for the last two cases, which shows that the amplitude will be doubled in the course of 24 hours. Thus we see that this kind of dynamic instability is at least of the right order of magnitude and may be one of the main factors responsible for the growth of disturbances in the upper atmosphere. For other cases, the amplifica-

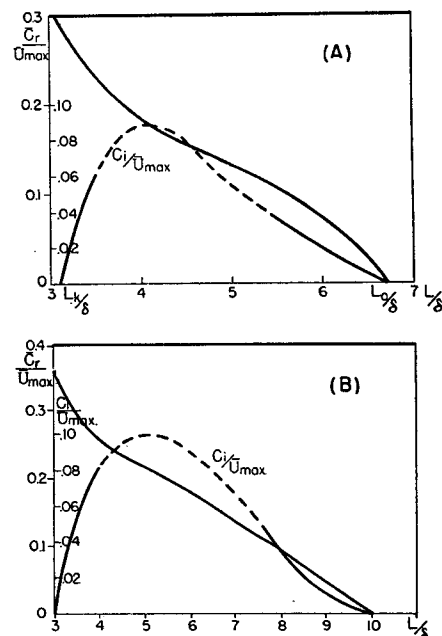


FIG. 11. c_r and c_i for the second sine-curve velocity profile. A: for the weaker jet; B: for the stronger jet, with the critical point at the same position.

tion factor may be much smaller, so that a much longer time would be required for the systems to double their amplitudes. Disturbances with a very small amplification factor will be difficult to distinguish from the stable waves. We have seen that for cases with very small amplification factors, the regions of instability are far away from each other and the value of U_* is very near to U_0 , *i.e.*, y_* is near the boundary; these relatively stable waves thus merge into the long stable waves.

8. The dynamic instability of geostrophic flow for three-dimensional perturbations in a baroclinic atmosphere

We shall discuss briefly the possibility of extending this study to more general conditions. Consider a stratified atmosphere, with both vertical and horizontal wind shear, and solenoidal field. The basic state is a geostrophic flow and hydrostatic equilibrium, defined by

$$fU = -S \frac{\partial P}{\partial y}, \quad g = -S \frac{\partial P}{\partial z}, \quad (61)$$

where S is the specific volume. From these we obtain the thermal wind relation

$$fU_z + g\sigma_y = fU\sigma_z, \quad (62)$$

where $\sigma \equiv \ln S$, and the subscripts denote differentiation; thus, σ_y and σ_z are the vertical and horizontal stratifications. We assume that the perturbation, represented by the small letters u , v , w , p , and s , is superposed on this basic flow. In the vertical direction, we assume that static equilibrium is still valid, *i.e.*, $dw/dt - f_z U$ can be neglected. Through this assumption, the very short gravitational waves are eliminated. Using the symbol L for the operator $\partial/\partial t + U\partial/\partial x$, the equations of motion and of continuity are then as follows:

$$Lu + (U_y - f)v + (U_z + f_z)w + Sp_x = 0, \quad (63)$$

$$Lv + fu - fUS^{-1}s + Sp_y = 0, \quad (64)$$

$$-gS^{-1}s + Sp_z = 0, \quad (65)$$

$$S^{-1}Ls + \sigma_y v + \sigma_z w - \text{div}(u, v, w) = 0. \quad (66)$$

The motion is assumed to be adiabatic, so that

$$S^{-1}Ls + \sigma_y v + \sigma_z w + S\epsilon(Lp + vp_y + wp_z) = 0, \quad (67)$$

where ϵ is the inverse of the Laplacian velocity of sound, and is taken as constant. From (63) and (64) we get the vorticity equation

$$L\zeta + (\beta - U_{yy})v + (f - U_y)\text{div}(u, v) - fUS^{-1}s_x - \sigma_y Sp_x = 0, \quad (68)$$

where the term $(f_z - U_z)\partial w/\partial y$ has been neglected. If we use the geostrophic relation for the meridional

velocity v ,

$$fv = Sp_x, \quad (69)$$

then by combining (65) and (69), we get

$$gs_x = fS(v_z - \sigma_z v). \quad (70)$$

Assuming that each of the perturbation quantities is expressed as a function of y and z multiplied by the harmonic factor $\exp[ik_x(x - ct)]$, and combining (67), (62), (69), and (70), we get ultimately

$$w = -rf(U - c)v_z - f(g^{-1}c - rU_z)v, \quad (71)$$

where $r^{-1} = g(\sigma_z - g\epsilon)$. From (66), (70), and (71), we get

$$\text{div}(u, v) = fr(U - c)(v_{zz} - \sigma_z v_z) + fg^{-1}Uv_z + [\sigma_y + fr(\sigma_z U_z - U_{zz}) - fg^{-1}U\sigma_z]v. \quad (72)$$

Substituting (69), (70), and (72) into (68), and assuming $\partial\zeta/\partial x = \nabla^2 v$, which is substantially equivalent to the geostrophic approximation for the vorticity, we find on neglecting smaller terms,

$$(U - c)[\nabla^2 v + rfZ(v_{zz} - \sigma_z v_z)] + [\beta - U_{yy} - \sigma_y U_y + rfZ(\sigma_z U_z - U_{zz})]v = 0, \quad (73)$$

where $Z = f - U_y$. A similar equation holds for the perturbation pressure p . It will be noted that, if we take the complete equations of motion without using the geostrophic relation, we arrive at a partial differential equation much more complicated than (73); but after neglecting terms of smaller magnitude by assuming that the waves are longer than 2000 km, we finally obtain (73).

Let us take the difference of the adiabatic and actual lapse rate equal to $3.5^\circ \text{C km}^{-1}$, and the mean temperature equal to 250°K ; then we shall have the following values:

$$(gr)^{-1} = \sigma_z - g\epsilon = 1.4 \times 10^{-5} \text{ m}^{-1}, \\ \sigma_z = 1.3 \times 10^{-4} \text{ m}^{-1}.$$

From (62),

$$f(\sigma_z U_z - U_{zz}) = g\sigma_y \sigma_z,$$

so the coefficient of the last term of (73) can be written

$$Z \frac{d}{dy} [\ln(SZ) + gr\sigma_z] - f\sigma_y,$$

if r is taken as constant. The terms in the expression above are all of the order $10^{-11} \text{ m}^{-1} \text{ sec}^{-1}$, the same as β . If we neglect the variation in the meridional direction, (73) reduces to the fundamental equation discussed by Charney (1947). This equation should be solved under certain boundary conditions, but it is so complicated that it is almost impossible to solve in this form. However, we may solve equations of this form by certain approximations. Thus if we neglect the variation of U with height when it is undifferentiated, but retain the terms with U_z , we may assume the exponential solution $v = V(y) \exp(\frac{1}{2}\sigma_z z + ik_z z)$;

(73) is then reduced to an ordinary differential equation with the independent variable y :

$$(U - c)\{V'' - [k_z^2 + rfZ(k_z^2 + \frac{1}{4}\sigma_z)]V\} + \{Z d[\ln(SZ) + g\sigma_z]/dy - f\sigma_y\}V = 0. \quad (74)$$

The function V satisfies the upper boundary condition that the energy per unit volume will be zero at infinity, but does not satisfy the lower condition $w = 0$. This condition could be satisfied by taking into consideration a small friction near the ground, an artifice used by Rayleigh. If k_z is real, we have internal waves, with the amplitude increasing with height; if, on the other hand, k_z is imaginary, we shall have external waves, with amplitude decreasing with height. This difference is determined by the wave length and by the value of $U - c$, which determines the relative values of the terms in the coefficient of V . Thus we see that for waves with $L \approx 5000$ km, we shall have internal waves, while for waves with $L \approx 2000$ km, we have external waves. With these differences, which will affect the value of k_z and therefore L , (74) is of the same form as (2), only with different constants as coefficients. Thus all the results we have obtained so far can also be applied to the case of three-dimensional flow in a baroclinic atmosphere, with the appropriate modifications for the position of the critical point and the value of the critical wave length. In this case, the critical point is characterized by the vanishing of the coefficient of the last term of (74), which, if we neglect $f\sigma_y$ against β , will occur when the quantity $\ln(SZ) + g\sigma_z$ has an extreme value. If σ_z is constant, the critical point is the point where SZ has its extreme value. When σ_y is large, $f\sigma_y$ will no longer be negligible compared with β .

9. Synoptic examples

In conclusion, we shall offer some speculations concerning the development of cyclone waves along a jet of westerlies. Suppose we have two critical points (where the absolute vorticity has an extreme value), one on each side of the maximum wind; then the air to the north will have an excess of (cyclonic) absolute vorticity relative to the air to the south. Thus if a disturbance sets in and an exchange of air results, cyclonic and anticyclonic circulations will be produced. If the two critical points are not very far away from each other, the degree of instability will be great and the air in the central part of the current will be subject to much larger displacements, since it has a better chance to move across the critical points. If these wave disturbances move with a velocity roughly equal to the mean of U_k and U_{min} , rapid development should be expected.

In order to test this theory, let us consider some actual cases. During the period 29–31 January 1947

a strong jet of westerlies developed over the United States. The velocity profile for January 29 was computed along the meridian from Bismarck, N. D., to Brownsville, Tex., at the 500-mb level. This profile and the corresponding distribution of absolute vorticity Z are shown in fig. 12. According to the criterion obtained from the assumption of conservation of momentum, the situation should be absolutely stable, except at the 300-mb level, where a narrow zone of instability exists ($\partial U/\partial y > f$). But if we examine the distribution of Z , we find that the condition $\partial Z/\partial y = 0$ is satisfied at two latitudes, one to the south of the jet, at 34°N and one to the north, at 43°N , which shows that unstable and neutral waves can exist. The flow pattern was practically parallel on January 28; on the 29th, a small disturbance developed. This small disturbance developed into a closed center on the 30th and was moving ENE with a speed of about 15 m sec^{-1} (12 degrees latitude per day), less than the wind velocity at the critical point, which is about 26 m sec^{-1}

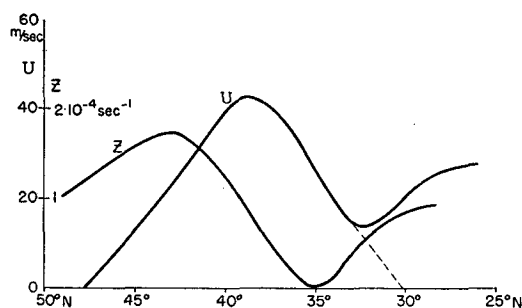


FIG. 12. The distribution of zonal velocity U and the absolute vorticity Z on the 500-mb surface for 29 January 1947. Note the symmetric position of the maximum and minimum Z in relation to the point with maximum current velocity.

(see fig. 13). Therefore, the cyclone was moving with a speed about one-half of U_k , corresponding roughly to a maximum amplifying factor. The dimension and intensity of this storm almost doubled in the course of one day and this trend of increase continued during the next 24 hours; this, at least qualitatively, agrees with our results.

The wind profile for this case is very similar to a sine curve, except that the critical point is nearer to the center of the belt, indicating a more favorable condition for instability. For a rough estimation of the wave length, we may use the results of section 7, taking $y_k = b/2$ so that $\alpha_k b = 2.7$. The observed width of the belt ($2b$) is about 15 degrees of latitude; hence, $L_k = 1920$ km. Since the observed speed of the cyclone is less than U_k and greater than U_{min} , the corresponding wave length must be around 2500 km to 3000 km.

The present theory can also be applied to some extent in explaining seasonal conditions. Since we can regard the activity of highs and lows as large-scale eddies, we should expect some instability in the region of frequent occurrence of lows and highs. Fig. 14 gives

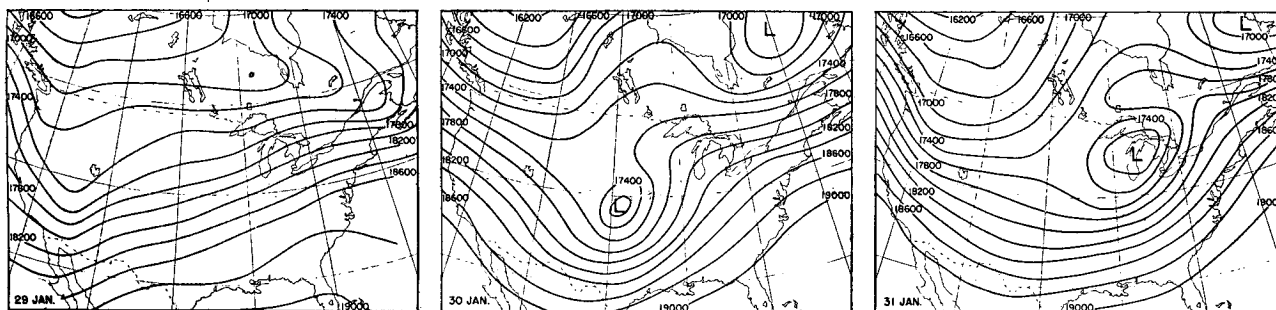


FIG. 13. 500-mb charts for 0300 GCT 29-31 January 1947. Note the formation and development of the cyclone center in the straight current and the rate of motion of this system.

the wind distribution at 12 km over North America during winter, which is plotted by using the values given by Willett (1944) and by Hess (1948). From the curve of Z , we see that there are two critical points, at about 40°N and 50°N . The region around 40°N to 50°N is actually the region of most frequent disturbances in winter. The curve for summer (fig. 14) shows that the region of instability has shifted northward to about 60°N . The vorticity profile is quite flat, indicating a lesser degree of instability according to the results we have obtained.

10. Summary

A general description of the method used in this study and the results obtained are presented in the following paragraphs.

Section 2 is concerned with the perturbation equations, the boundary conditions, and the types of velocity profiles. The study is mainly for horizontal nondivergent wave motions, so that the basic equation to be solved is the vorticity equation, which, when the harmonic wave solution is assumed, reduces to a differential equation for the stream function or the meridional velocity v . Since the velocity profiles of the westerlies generally have some symmetry with respect to a certain latitude, only symmetric profiles and boundary conditions are discussed. This results in symmetry of the perturbation, and leads to the simplification that only half of the profile need be considered. At the midpoint of the belt, where the maximum wind

velocity is found, the perturbation should have its maximum value. Two different kinds of profiles are discussed: the first is a symmetric profile in a limited belt, with the boundary conditions that the perturbation vanishes at two latitudes $y = y_1$ and $y = y_2$; the second type is a belt extending to infinity in both directions so that the perturbation will be zero only at infinity.

In section 3 the analytic properties of the stream function are discussed. In this study our attention is centered on the range of phase velocities with a value between the maximum and minimum current velocity in the belt, so that we always have the point where $c = U$, which is a *singular point* of the vorticity equation, where the general solution ceases to be valid. Heretofore, meteorologists have always avoided this by assuming $c < U$; but by so doing have restricted their investigations to the slowly moving or retrograding waves.

In the immediate vicinity of this singular point (where $c = U$) the solution will give an infinite perturbation velocity and a sudden change of phase which can never occur in the real atmosphere. This difficulty is obviated by taking account of viscous (or eddy) stresses, which may be negligible as compared with the other effects when c is quite different from U , but must be of principal importance for the region in the vicinity of the singular point where $c = U$, and can not be neglected there. When the terms due to lateral mixing are included, the point where $c = U$ is no longer a singular point of the vorticity equation and infinite

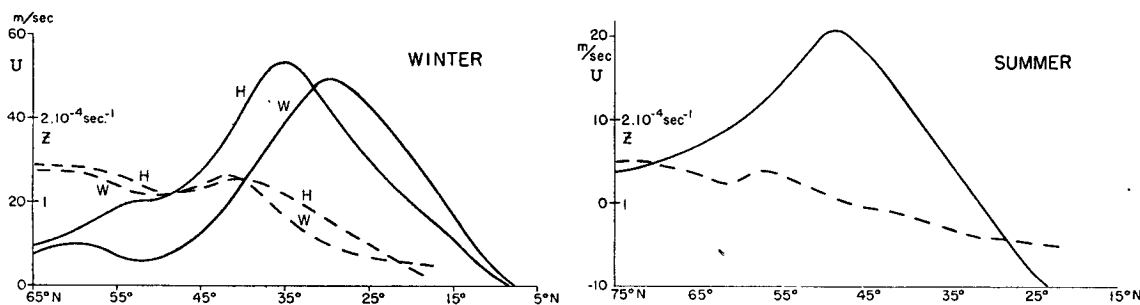


FIG. 14. Seasonal mean of meridional distribution of zonal wind velocity (full lines) and absolute vorticity (broken lines) after Willett (W) and Hess (H). Note the flatness of the vorticity curve in summer.

perturbation velocities do not occur. The change of phase will also be a gradual one. At a finite distance from the point where $c = U$, the frictional terms can be neglected entirely and the motion will be represented adequately by the solution of the ordinary vorticity equation. Thus it is only necessary to correct our results for the immediate vicinity of the singular point by considering the effect of virtual viscosity; this is the so-called method of cross-substitution.

Section 4 is devoted to the neutral waves (waves with amplitude independent of time). It is found that the phase velocity of these neutral waves can never be greater than the maximum wind velocity within the belt, but unlike the corresponding waves in a non-rotating system, they may have a phase velocity less than the minimum wind velocity, examples of which are the waves represented by Rossby's or Haurwitz' formula for the case with $U = \text{constant}$. When c has a value between the maximum and minimum current velocities, the occurrence of neutral waves requires that the absolute vorticity possess an extreme value ($dZ/dy \equiv \beta - U'' = 0$) at some point. If no such *critical point* exists in the profile, then there can be no neutral wave with a phase velocity in the range between minimum and maximum current velocities. If only one distinct point (the two symmetric points, one on each side of the midpoint of the profile are identical) with this property exists, then only one such neutral wave will be possible, and its velocity will be given by the value of $U (= U_k)$ at this point. Thus, in order to find the phase velocity of the neutral waves within this range of values, we need only plot the distribution of the absolute vorticity $Z = f - U'$ and find the point where $dZ/dy = 0$; the phase velocity is given by the current velocity at this point. The constant-vorticity profile is a special case for which this condition is satisfied at every point; in this case, all waves are neutral.

In section 5 it is found that the necessary condition for having amplified (unstable) waves is that the absolute vorticity Z must have an extreme value within the belt, *i.e.*, $dZ/dy \equiv \beta - U''$ must be zero at some point $y = y_k$ (the critical point) in the belt, and change its sign there. The same conclusion also holds for motion over a spherical earth. If no such critical point exists, then all waves with phase velocity greater than the minimum current velocity will be damped.

Section 6 deals with waves having a phase velocity slightly different from that of the neutral wave (with phase velocity $c = U_k$ and wave length $L = L_k$), or slightly greater than $U_{\min}$. It is found that for waves longer than L_k , the phase velocity c_r (the real part of c) will be smaller than U_k and c_i (the imaginary part of c) will be positive, *i.e.*, their amplitude will increase with time, while for waves shorter than L_k , we shall have $c_r > U_k$ and $c_i < 0$, *i.e.*, the shorter and fast

moving waves are damped. Thus, the neutral wave with phase velocity equal to the current velocity at the critical point divides the disturbances into amplified and damped waves. Equations are developed for calculating c_r and c_i in the neighborhood of this neutral solution.

The waves with a phase velocity slightly greater than that of the neutral wave with $c = U_{\min}$ are shown to be shorter and amplified.

Since the waves with $c = U_k$ and $c = U_{\min}$ are the only neutral waves with a phase velocity within the range of current velocities, all waves with a value of the phase velocity between these two values will be amplified. Thus, a diagram with L as abscissa and c_r (or c_i) as ordinate is divided into three parts by the points $P(L_0, c_0 = U_{\min})$ and $Q(L_k, c_k = U_k)$; to the right of P , with $L > L_0$, the waves are essentially neutral, with $c < U_{\min}$; to the left of Q , where $L < L_k$ and $c_r > U_k$, we shall have damped waves; in the central part, the waves are all amplified. The greatest degree of amplification will occur when c_r has a value somewhere between c_0 and c_k , and the sharper the velocity profile the greater the instability; both fast and slowly moving waves will show little amplification.

When the waves are amplified, the troughs and ridges will not be straight lines but will be directed from southwest toward northeast to the south of the point where the absolute vorticity has its minimum value and from southeast toward northwest to the north of the point of maximum vorticity. This kind of flow pattern will give a negative correlation between the perturbation velocities u and v , and therefore indicates instability.

Section 7 is devoted to a study of numerical examples with sine-curve profiles. The wind distribution is fitted by the parameters $U_{\max}$, the width of the belt $2b$, and the position of the critical point. All the results obtained in section 6 are substantiated by these periodic profiles. The order of magnitude of c_i is such that a doubling of amplitude in the course of 24 hours could be expected in some cases.

In section 8, the study is extended to three-dimensional motions in a baroclinic atmosphere, and it is found that if certain approximations are made (which are justifiable for large-scale atmospheric motions), a second-order differential equation will be obtained for the meridional perturbation velocity v (also satisfied by the perturbation pressure), which is of the same form as the vorticity equation for two-dimensional motions, but with different coefficients. In this case, the critical point is determined not by the extreme value of the absolute vorticity Z , but by that of $\ln(SZ) + g\sigma_z$, where S is the specific volume, σ_z the vertical stratification, and r a quantity depending on σ_z . Thus all the results obtained remain valid provided

some modifications as to the values of c_k , L_k , and the other quantities are made.

In section 9, a synoptic example for the period of 28–31 January 1947 is studied. The situation was such that unstable waves could be expected. A cyclone center developed over the United States and moved with a velocity which, theoretically, indicated maximum amplification. Deepening and intensification of the cyclone actually took place in the following few days.

Acknowledgments.—The writer wishes to express his gratitude to Prof. C.-G. Rossby for his guidance and help during the course of this study, to Prof. Zd. Sekera, Prof. C. C. Lin, and T. C. Yeh for suggestions and discussions.

APPENDIX

A. The amplified waves in the vicinity of the neutral solution at the critical point.—In (11), $\Psi_1(0; \alpha, c)$, $\Psi_2(0; \alpha, c)$, $\Psi_1'(b; \alpha, c)$, and $\Psi_2'(b; \alpha, c)$ are all analytic functions of α and c ($y = 0$ is the boundary, $y = b$ the midpoint); therefore, $F(\alpha, c)$ is also analytic and we can continue the solution for c in terms of $\lambda \equiv \alpha^2$ in the neighborhood of $\alpha = \alpha_k$, $c = c_k = U_k$, and write

$$c = c_k + \left(\frac{dc}{d\lambda} \right)_k (\lambda - \lambda_k) + \frac{1}{2} \left(\frac{d^2c}{d\lambda^2} \right)_k (\lambda - \lambda_k)^2 + \dots \quad (75)$$

The problem is then to find the differential coefficients of c with respect to λ in the vicinity of $\lambda = \lambda_k$.

For this purpose, we write the Sturm-Liouville system in the form

$$\Psi'' - \left(\lambda + \frac{U'' - \beta}{U - c} \right) \Psi = 0; \quad (76)$$

$$\Psi(0) = 0 = \Psi'(b),$$

which has one characteristic solution $\Psi = \Psi_k$ corresponding to $c = c_k = U_k$, $\lambda = \lambda_k \equiv \alpha_k^2$. If we have another solution $\Psi = \Psi_k + (d\Psi/d\lambda)_k \Delta\lambda$, which is slightly different from Ψ_k , and if $c_k + (dc/d\lambda)_k \Delta\lambda$ is the corresponding value of c , then we find from (76) that Ψ_k and $(d\Psi/d\lambda)_k$ must satisfy the equations

$$\Psi_k'' - \left(\lambda_k + \frac{U'' - \beta}{U - c_k} \right) \Psi_k = 0,$$

$$\left(\frac{d\Psi''}{d\lambda} \right)_k - \left(\lambda_k + \frac{U'' - \beta}{U - c_k} \right) \left(\frac{d\Psi}{d\lambda} \right)_k = \left[1 + \frac{U'' - \beta}{(U - c_k)^2} \left(\frac{dc}{d\lambda} \right)_k \right] \Psi_k.$$

Multiply the first by $(d\Psi/d\lambda)_k$ and the second by Ψ_k

and subtract; since both Ψ_k and $(d\Psi/d\lambda)_k$ satisfy the boundary conditions, we have, on integrating the result from 0 to b , the following condition

$$\int_0^b \left[1 + \frac{U'' - \beta}{(U - c_k)^2} \left(\frac{dc}{d\lambda} \right)_k \right] \Psi_k^2 dy = 0, \quad (77)$$

or

$$\Delta\lambda \int_0^b \Psi_k^2 dy + \Delta c \left(E + \int_{y_k-\epsilon}^{y_k+\epsilon} \frac{U'' - \beta}{(U - c_k)^2} \Psi_k^2 dy \right) = 0, \quad (78)$$

where

$$E \equiv \lim_{\epsilon \rightarrow 0} \left(\int_0^{y_k-\epsilon} + \int_{y_k+\epsilon}^b \right) \frac{U'' - \beta}{(U - c_k)^2} \Psi_k^2 dy. \quad (79)$$

Suppose that $U_k''' - \beta' \neq 0$ at $y = y_k$, so that $y = y_k$ is a simple root of $U''(y_k) - \beta = 0$, and take $\eta \equiv y - y_k$; then, in the neighborhood of $\eta = 0$,

$$\begin{aligned} \Psi_k &\approx 1 \\ U - U_k &= U - c_k \approx U_k' \eta \\ U'' - \beta &\approx U_k''' \eta. \end{aligned}$$

With these values for the vicinity of $y = y_k$, and with $D \equiv (U_k''' - \beta')/U_k'^2$, we find

$$\int_{y_k-\epsilon}^{y_k+\epsilon} \frac{U'' - \beta}{(U - c_k)^2} \Psi_k^2 dy = D\pi i. \quad (80)$$

Writing $\Delta c = \Delta c_r + ic_i$, and using (80), equation (78) may be solved for Δc_r and c_i ; by this process we are led to equations (32) and (33). As has been discussed in section 6, c_i is always positive for $\Delta\lambda < 0$ and negative for $\Delta\lambda > 0$, if $D < 0$, which is generally the case.

For the second type of profile we may apply the same reasoning as for the first, but now we must integrate from zero to $y = -\infty$ instead of $y = \delta$, for although we can use the condition $\Psi' + r\Psi = 0$ at the point of transition, this leaves $(d\Psi/d\lambda)_k$ in the equation, which cannot be taken as Ψ_k . If we integrate (77) from $y = -\infty$ to $y = \delta$ (taking the origin at the point of transition) and use the same procedure as before, we shall find that c_i and Δc_r are given by the same expressions (32) and (33), except that E is replaced by

$$E - a^2\beta/2r(U_k - U_0)^2$$

and $\int_0^b \Psi_k^2 dy$ by

$$\int_0^\delta \Psi_{k1}^2 dy + \frac{a^2}{2r},$$

where Ψ_{k1} is the neutral solution for the central part of the profile and a its value at the point of transition $y = 0$.

B. The amplified waves in the vicinity of the stationary neutral solution.—For $U(0) = 0$, (16) reduces to

$$c \left(\Psi'(0) + \alpha^2 \int_0^b \Psi dy \right) + \beta \int_0^b \Psi dy - \alpha^2 \int_0^b \Psi U dy = 0, \quad (81)$$

which is true for all values of c ; the problem is to find Ψ . Let us differentiate (81) with respect to $\lambda \equiv \alpha^2$ in the neighborhood of $\Psi_{c=0}$:

$$\left(\frac{dc}{d\lambda} \right)_0 \left(\Psi'_0(0) + \alpha_0^2 \int_0^b \Psi_0 dy \right) + \beta \int_0^b \left(\frac{d\Psi}{d\lambda} \right)_0 dy - \int_0^b U \Psi_0 dy - \alpha_0^2 \int_0^b U \left(\frac{d\Psi}{d\lambda} \right)_0 dy = 0, \quad (82)$$

where Ψ_0 and α_0^2 are the values for $c = 0$ and therefore satisfy

$$\beta \int_0^b \Psi_0 dy = \alpha_0^2 \int_0^b U \Psi_0 dy. \quad (38)$$

In order to evaluate $(dc/d\lambda)_0$ in (82), we have to find the expressions for Ψ_0 and $(d\Psi/d\lambda)_0$. Let us consider the point very near to $c = 0$, with $c_r > 0$ and $|c_i| \ll c_r$. Take the new origin at the singular point where $U = c$ in the complex plane; expand U as a Taylor series with respect to the point $U = c$ (see fig. 3). Evaluating this series at the point $y = y_0$ where $U = 0$, we have

$$0 = c + U'_s y_0 + \dots$$

Reverting,

$$y_0 = -c/U'_s + \dots;$$

also,

$$\begin{aligned} U'_0 &= U'_s + U''_s y_0 + \dots \\ &= U'_s - U''_s c/U'_s + \dots \end{aligned}$$

By making use of the condition that $\Psi = 0$ at the boundary $y = y_0$, the ratio of the two arbitrary constants in $\Psi = A\Psi_1 + B\Psi_2$ is found to be

$$\begin{aligned} \frac{A}{B} &= -\frac{\Psi_2(y_0)}{\Psi_1(y_0)} \\ &= \frac{1}{c} \left[U'_s - \frac{U''_s - \beta}{U'_s} c \ln \left(-\frac{c}{U'_s} \right) + \dots \right]. \end{aligned}$$

Thus we may write, with $G_s \equiv (U''_s - \beta)/U'_s$,

$$\begin{aligned} \Psi &= B_1 \left\{ \left[U'_s - cG_s \ln \left(-\frac{c}{U'_s} \right) + \dots \right] \Psi_1 \right. \\ &\quad \left. + c(1 + G_s y \ln y + \dots) \right\}. \quad (84) \end{aligned}$$

Since the imaginary part of $\ln(-c/U'_s)$ is $-\pi i$ and that of $y \ln y$ and all other terms is small for small c_r

and c_i , the main imaginary part of Ψ is, if we take $B_1 = 1$,

$$\Psi_{i0} \approx c_r G_s \Psi_1 \pi i \approx c_r G_s U(y) \pi i / U'_0. \quad (85)$$

If we assume that (83) is also satisfied by $(d\Psi/d\lambda)_0$ and take $U(y)$ as the approximate expression for Ψ_{r0} , then, after separating the real and imaginary parts of (82), we get

$$K_r c_r - K_i c_i = \Delta(\alpha^2) \int_0^b U^2 dy, \quad (86)$$

$$K_r c_i + K_i c_r = \Delta(\alpha^2) \int_0^b U \Psi_{i0} dy, \quad (87)$$

where

$$K_r = U'(0) + \alpha_0^2 \int_0^b U dy, \quad (88)$$

$$K_i = \alpha_0^2 \int_0^b \Psi_{i0} dy. \quad (89)$$

In (89) we have used the condition $\Psi'_{i0} = 0$; further, on reference to (85), it is seen that K_i is of the order of c_r . Hence, since $c_i \ll c_r$, we may neglect $K_i c_i$ against $K_r c_r$ in (86), which then reduces to (35). Finally, the result given in (36) is obtained by substitution of (35) and (85) in (87).

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